Lorentzian Endpoint Conditions Do Not Select the Einstein Sector of Pure Weyl Gravity

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Abstract

Maldacena-type boundary selection isolates an Einstein sector of conformal gravity in Euclidean anti-de Sitter or late-time de Sitter settings. We ask whether physically familiar one-ended Lorentzian conditions do the same for linear perturbations of a Schwarzschild black hole in four-dimensional pure Weyl gravity. They do not.

In the axial $\ell = 2$ sector we derive the linearized Bach system from the Weyl action and identify an exact filtered differential module with successive quotients

$$M^{(2)}_{\mathrm{RW}}, \qquad M^{(2)}_{\mathrm{RW}}, \qquad M^{(1)}_{\mathrm{RW}}.$$

The two spin-two factors form a non-split self-extension. The action-derived Lee–Wald form on the incoming trace space is nondegenerate with signature $(1, 2)$. In factor coordinates it is congruent, for every real $\omega > 0$, to

$$\pi\alpha_{\mathrm{W}} \begin{pmatrix} 0 & 576\omega/5 & 0 \\ 576\omega/5 & 0 & 0 \\ 0 & 0 & -32/(15\omega) \end{pmatrix}.$$

Scalar Regge–Wheeler Wronskians and boundary dévissage prove that the map from future-horizon-regular data to this complete incoming Krein space is invertible for every positive real frequency. Negative-flux directions are therefore populated by regular exterior solutions, rather than being formal endpoint artefacts.

On the certified cell $\omega \in [0.49995, 0.50005]$ the outgoing and future-horizon spaces are matching three-channel Krein spaces, the outgoing map is invertible, and the one-sided scattering map obeys the exact integrated pseudo-isometry. Analytic continuation gives outgoing population on an open dense, full-measure subset of the positive axis; possible failures are isolated scalar reflection zeros. The same filtration excludes exponentially growing separated axial modes.

Finally, a validated projective Evans calculation encloses one simple damped spin-two quasinormal frequency and certifies a nonzero intrinsic extension selector. The spin-one factor is a local unit there. The complete connection therefore has Smith valuations $(0, 0, 2)$, algebraic multiplicity two, geometric multiplicity one, and one length-two connection root chain. This is a connection-level intrinsic exceptional point. A second-order physical Green-resolvent pole remains conditional on a separate analytic Fredholm realization.

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1 Question, answer, and scope

Pure Weyl gravity is defined by

$$S_{\mathrm{W}}[g] = \alpha_{\mathrm{W}} \int_M \sqrt{-g}\, C_{abcd}C^{abcd}\, \mathrm{d}^4x, \qquad B_{ab} = 0. \tag{1}$$

Every Einstein metric is Bach-flat, but the fourth-order Bach equation has additional local solutions. The question addressed here is deliberately narrow:

> Do future-horizon regularity and the ordinary incoming or outgoing real-frequency conditions at null infinity force a Schwarzschild perturbation of pure Weyl gravity into the linearized Einstein kernel?

For the complete axial $\ell = 2$ system the answer is no. The extra curvature carrier survives the endpoint conditions, participates in the action-derived current, and supplies populated negative directions in a nondegenerate Krein space.

This is not a statement that no boundary condition can select the Einstein sector. The local Cauchy restriction

$$\delta R_{ab}|_\Sigma = 0, \qquad \nabla_n \delta R_{ab}|_\Sigma = 0$$

does select the linearized Einstein kernel in the axial carrier problem. Nor is this paper a quantum-positivity theorem. "Negative" below always means negative with respect to the classical Lee–Wald flux form.

1.1 Five principal results

The paper is organized around five statements.

1. The complete axial system has an exact RW/RW/spin-one filtration, with a nontrivial repeated-spin-two extension.
2. Its incoming action-flux space has signature $(1, 2)$ and is populated at every $\omega > 0$ by future-horizon-regular solutions.
3. On a certified nonzero frequency cell the complete one-sided scattering map is pseudo-isometric between matching Krein spaces; outgoing population is generic on the positive real axis.
4. The full filtered system has no exponentially growing separated axial mode.
5. One enclosed damped spin-two quasinormal frequency is a certified connection-level exceptional point with Smith valuations $(0, 0, 2)$.

1.2 Controlled vocabulary

Definition 1.1. The *Einstein kernel* is the kernel of the linearized Ricci map $h \mapsto \delta R_{ab}[h]$. The *Ricci carrier* is its image on linearized Bach solutions. *Population* of an endpoint trace space means surjectivity of the relevant horizon-to-endpoint connection map. *Factor frame* means a basis adapted to the differential-module filtration. A *signature frame* diagonalizes a Hermitian flux form. These two frames are not identified.

The result tags used below have the repository meanings LOCAL-ALGEBRAIC, REDUCED-MODE, and LORENTZIAN-CAUSAL. A reduced-mode theorem is never used as evidence for a complete Lorentzian quantum or time-domain construction.

2 Relation to existing work

Maldacena's conformal-gravity construction uses a Neumann condition in Euclidean AdS or at late times in de Sitter space to select Einstein solutions [1]. The present problem has a different signature, background, and boundary geometry: a one-ended Lorentzian Schwarzschild exterior with a future event horizon and null infinity.

Ricci-tensor variables have long been useful for exposing higher-derivative spin-two sectors. Recent quadratic-gravity work computes axial quasinormal modes of massless and massive spin-two fields [5], develops a Ricci-tensor treatment of Kerr perturbations [6], evolves Einstein–Weyl perturbations in time [7], and studies the excitation and suppression of beyond-GR ringdown by point-particle sources [8]. A recent thesis gives a complementary radial Ricci-tensor reconstruction and its relation to the Schwarzschild–Bach family [9].

Those works concern quadratic or Einstein–Weyl theories, massive sectors, radial reconstructions, or observable excitation. Our subject is the degenerate massless self-extension in strict pure C^2 gravity and the indefinite current derived from that action. The claims are therefore complementary: we do not infer excitation amplitudes or a complete waveform from endpoint population alone.

3 Ricci–Bach composition

Let the background be Ricci-flat. For a metric perturbation h_{ab}, put

$$\psi_{ab} = \delta R_{ab}[h], \qquad \psi = g^{ab}\psi_{ab}.$$

Linearization of the Bach tensor gives

$$\delta B_{ab}[h] = \frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} - \frac{1}{6}\nabla_a\nabla_b\psi - \frac{1}{12}g_{ab}\Box\psi. \tag{2}$$

In the axial sector $\psi = 0$, so the carrier equation is a second-order Lichnerowicz-type equation

$$L_{\mathrm{car}}\psi_{ab} := \frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} = 0. \tag{3}$$

Proposition 3.1 (Ricci–Bach exact sequence)**.** *On the declared axial Schwarzschild $\ell = 2$ mode class, the linearized Ricci map fits into the exact sequence*

$$0 \longrightarrow \mathcal{E}_{\mathrm{RW}} \longrightarrow \mathcal{B}_{\mathrm{axial}} \xrightarrow{\delta\,\mathrm{Ric}} \mathcal{C}_{\mathrm{axial}} \longrightarrow 0, \tag{4}$$

where $\mathcal{E}_{\mathrm{RW}}$ is the two-dimensional Regge–Wheeler Einstein module and $\mathcal{C}_{\mathrm{axial}}$ is the realized second-order Ricci carrier. The sequence is exact but has no certified canonical metric splitting.

Proof sketch. Equation (2) proves that the Ricci image of a Bach solution obeys (3). The complete reconstruction audit verifies every Ricci row, including the independent $v\varphi$ equation omitted in an earlier shallow reduction. Its zero fibre is exactly the linearized Einstein Regge–Wheeler system. Exact recurrence and reconstruction certificates show that the declared carrier is realized. A rational splitting is separately obstructed by the extension invariants discussed in Section 4. $\square$

3.1 Horizon reach

In ingoing Eddington–Finkelstein coordinates the carrier horizon is a regular singular point. For every nonzero real frequency its ingoing analytic solution space is two-dimensional. Thus horizon regularity alone does not imply $\psi_{ab} = 0$.

Corollary 3.2 (Local horizon nonselection)**.** *Future-horizon analyticity admits nonzero Ricci-carrier data in addition to the Einstein kernel. Hence it does not select the Einstein image within the axial Bach solution space.*

4 The axial filtration and its partial jet

After exact factor reduction the six-state axial system has a filtration

$$0 \subset F_1 \subset F_2 \subset F_3 = M_{\text{axial}},$$

with

$$F_1 \simeq M_{\text{RW}}^{(2)}, \qquad F_2/F_1 \simeq M_{\text{RW}}^{(2)}, \qquad F_3/F_2 \simeq M_{\text{RW}}^{(1)}. \tag{5}$$

Here $M_{\text{RW}}^{(2)}$ is the axial spin-two Regge–Wheeler module and $M_{\text{RW}}^{(1)}$ is the spin-one quotient with potential

$$V_1(r) = \frac{6(r-2)}{r^3}$$

in units $M = 1$.

In factor variables $X, Y, Z \in \mathbb{C}^2$, the generator is

$$\begin{aligned} X' &= AX + EY + CZ, \\ Y' &= AY + DZ, \\ Z' &= A_1 Z. \end{aligned} \tag{6}$$

The two upper sources share one image line:

$$E = USJ, \qquad C = USN.$$

Consequently (6) is exactly the spin-two-row first jet of the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}. \tag{7}$$

Over the dual numbers $\mathbb{C}[\varepsilon]/(\varepsilon^2)$, the six-state solution is simply $(Y + \varepsilon X, Z)$.

Theorem 4.1 (Exact axial partial-jet filtration)**.** *Every interior transport matrix has the form*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \tag{8}$$

where

$$\begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}$$

is the transport of (7)*. In particular*

$$\det \Phi_6 = (\det P)^2 \det R.$$

The repeated-spin-two extension is non-split over the generic rational differential field and is not generated, modulo rational gauge, by frequency, Schwarzschild-mass, or angular-eigenvalue differentiation.

The jet identity is useful both conceptually and numerically. A triangular connection

$$T=\begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & f \end{pmatrix}$$

is the jet of

$$C(\tau)=\begin{pmatrix} a(\tau) & d(\tau) \\ 0 & f \end{pmatrix}, \quad \dot{a}=b, \quad \dot{d}=c.$$

Therefore

$$T^{-1}=\begin{pmatrix} a^{-1} & -b/a^2 & (bd-ac)/(a^2f) \\ 0 & a^{-1} & -d/(af) \\ 0 & 0 & f^{-1} \end{pmatrix}, \tag{9}$$

and the three apparently independent extension ratios are one base transfer and its intrinsic derivative.

4.1 A local branch-resolving C operator is obstructed

The non-split extension also constrains possible positive-metric constructions. The statement is local and classical; it is not a theorem about a quantum Hilbert space.

Proposition 4.2 (No branch-resolving rational involution)**.** *Let*

$$0 \longrightarrow M_{\mathrm{RW}} \longrightarrow E_{\mathrm{Bach}} \longrightarrow M_{\mathrm{RW}} \longrightarrow 0$$

be the generic repeated-spin-two extension. There is no rational differential-module endomorphism C with $C^2=1$ that acts by $+1$ on the Einstein submodule and by -1 on the quotient.

Proof. In a triangular frame,

$$\mathcal{A}=\begin{pmatrix} A & E \\ 0 & A \end{pmatrix}.$$

A filtration-compatible involution with the declared actions must be

$$C=\begin{pmatrix} I & Q \\ 0 & -I \end{pmatrix}.$$

Covariant constancy,

$$C'+C\mathcal{A}-\mathcal{A}C=0,$$

requires

$$Q'+QA-AQ=-2E.$$

Over characteristic zero this says that the extension cocycle is a rational coboundary. It would split the extension, contradicting the certified nonzero rational extension class. □

There is an independent endpoint obstruction to a branch-preserving positive fundamental symmetry. The Einstein spin-two vector E is null. If a fundamental symmetry preserved its line, $CE=\pm E$, then the proposed positive majorant would satisfy

$$G(E,CE)=\pm G(E,E)=0.$$

Thus any positive fundamental symmetry must mix the two spin-two null directions. The elementary endpoint symmetry that exchanges them is a perfectly valid classical Krein fundamental symmetry, but it is not Mannheim's dynamical C operator. Proposition 4.2 shows that this distinction is structural: a dynamical C, if it exists, cannot be a local rational operator that resolves the two layers with opposite signs.

5 Action-derived endpoint flux

The sphere-integrated Lee–Wald current is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha_{\mathrm{W}} \int_{S^2} \sqrt{q}\,\big(h_B^{ab}\delta B_{ab}[h_A] - h_A^{ab}\delta B_{ab}[h_B]\big). \tag{10}$$

It is therefore not an arbitrarily normalized Wronskian of a chosen master field.

Let $G_-(\omega)$ be the incoming null-infinity Gram in the three-dimensional factor trace space. Define

$$E = EI0, \qquad R = XI0 - \frac{i}{\omega} XI1,$$

and let S be the unique G_--orthogonal lift of the unit spin-one quotient. After the harmless Einstein shear making the second spin-two factor null, the exact factor Gram is

$$\widehat{G}_-^{\mathrm{fac}} := \frac{G_-}{\pi\alpha_{\mathrm{W}}} = \begin{pmatrix} 0 & \frac{576}{5}\omega & 0 \\ \frac{576}{5}\omega & 0 & 0 \\ 0 & 0 & -\frac{32}{15\omega} \end{pmatrix}. \tag{11}$$

Theorem 5.1 (Canonical incoming Krein anatomy)**.** *For every real $\omega > 0$,*

$$\operatorname{inertia} G_-(\omega) = (1, 2, 0).$$

The repeated spin-two extension is a hyperbolic plane and the intrinsic spin-one quotient metric is strictly negative:

$$g_1 = -\pi\alpha_{\mathrm{W}} \frac{32}{15\omega}.$$

In normalized channels

$$P_2 = \frac{E+R}{\sqrt{2a}}, \qquad N_2 = \frac{E-R}{\sqrt{2a}}, \qquad N_1 = \frac{S}{\sqrt{b}},$$

where $a = 576\omega/5$ and $b = 32/(15\omega)$, the Gram is

$$\mathrm{diag}(1, -1, -1).$$

The positive Hilbert majorant associated with this Krein form is

$$\|(e, r, s)\|_{\mathrm{maj}}^2 = \int_0^\infty \left[\frac{576}{5}\omega(|e|^2 + |r|^2) + \frac{32}{15\omega}|s|^2\right] \mathrm{d}\omega. \tag{12}$$

Thus the natural time-domain exponents are

$$(e, r) \in \dot{H}^{1/2}, \qquad s \in \dot{H}^{-1/2}.$$

This identifies the canonical completion but does not itself prove global boundedness of the scattering multiplier.

6 Every incoming channel is populated

Let $T_-(\omega)$ map future-horizon-regular coefficients to incoming null-infinity factor traces. Its factor form is triangular:

$$T_- = \begin{pmatrix} a_2 & b & c \\ 0 & a_2 & d \\ 0 & 0 & a_1 \end{pmatrix}, \qquad a_s = A_{\mathrm{in},s}. \tag{13}$$

Lemma 6.1 (Boundary dévissage)**.** *Suppose a filtered differential system has boundary-compatible inclusion and quotient maps, and none of its scalar quotient problems admits a nonzero solution in the declared boundary class. Then the full filtered system has no nonzero solution in that class.*

Proof. Project a full solution to the top quotient. The quotient boundary problem forces the projection to vanish, so the solution lies in the preceding submodule. Repeat through the filtration. The last scalar factor then forces the remaining solution to vanish. No diagonalization or splitting of the extension is required. □

For either scalar Regge–Wheeler factor, real-frequency current conservation gives

$$|A_{\mathrm{in},s}|^2 - |A_{\mathrm{out},s}|^2 = 1. \tag{14}$$

Thus $A_{\mathrm{in},s} \neq 0$ for every real $\omega > 0$.

Theorem 6.2 (Global incoming population)**.** *For the complete axial $\ell = 2$ system,*

$$T_-(\omega) \in GL(3, \mathbb{C}) \qquad \textit{for every real } \omega > 0.$$

Consequently every positive, negative, and null vector of the incoming Krein space is the trace of a unique future-horizon-regular solution.

Proof. A vector in $\ker T_-$ would be horizon regular and purely outgoing at infinity. Its spin-one projection vanishes by (14). Boundary dévissage then removes the carrier and metric spin-two factors successively. Hence the kernel is zero. Since T_- is a 3×3 map, it is invertible. □

The conclusion is stronger than existence of formal negative directions: the entire indefinite incoming space is physically reached within the declared exterior boundary problem.

7 Outgoing population and one-sided Krein scattering

Let T_+ be the future-null-infinity outgoing connection and $H_{\mathcal{H}^+}$ the outward-oriented future-horizon Gram. Conservation of (10) gives

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0. \tag{15}$$

The explicit entries of T_+ are not needed to decide population on the certified cell. The scalar outgoing factors and the exact filtration give

$$\det T_+ = \rho_+ A_{\mathrm{out},2}^2 A_{\mathrm{out},1}, \qquad \rho_+ \neq 0.$$

Theorem 7.1 (Certified outgoing cell)**.** *For $M = 1$, axial $\ell = 2$, and*

$$I_0 = [0.49995, 0.50005],$$

validated endpoint calculations prove

$$|A_{\mathrm{out},2}| > 0.011326240435330133, \qquad |A_{\mathrm{out},1}| > 0.14680548488890086.$$

Hence $T_+(\omega)$ is invertible throughout I_0. On the exact common cell

$$I_* = [1/2, 10001/20000],$$

the incoming, outgoing, and future-horizon coefficient spaces are nondegenerate three-channel Krein spaces of inertia $(1, 2, 0)$.

In incoming coordinates define

$$R = T_+T_-^{-1}, \qquad A = T_-^{-1}.$$

Multiplying (15) by $T_-^{-\dagger}$ and T_-^{-1} gives

$$G_- = R^\dagger G_+ R + A^\dagger H_{\mathcal{H}^+} A. \tag{16}$$

Equivalently,

$$\mathscr{S} = \begin{pmatrix} R \\ A \end{pmatrix}$$

is a pseudo-isometric embedding

$$(\mathbb{C}^3, G_-) \hookrightarrow (\mathbb{C}^3 \oplus \mathbb{C}^3, G_+ \oplus H_{\mathcal{H}^+}).$$

Corollary 7.2 (Band-limited pseudo-isometry)**.** *Multiplication by T_+ is a bounded isomorphism on $L^2(I_*; \mathbb{C}^3)$. Equation (16) therefore integrates to an exact pseudo-isometry of the complete band-limited wave-packet spaces.*

Corollary 7.3 (Generic positive-real outgoing population)**.** *Assume the factor-normalized scalar Jost coefficients are holomorphic away from the separate threshold $\omega = 0$. Their strict nonvanishing on I_0 proves that neither is identically zero. Therefore the positive-real exceptional set is locally finite and*

$$T_+(\omega) \in GL(3, \mathbb{C})$$

on an open dense, full-measure subset of $(0, \infty)$. On every compact positive-frequency band the corresponding multiplication operator is injective with dense range. It is a bounded isomorphism exactly when the band contains no exceptional frequency.

This does not prove that the exceptional set is empty. The next scattering experiment is to compute the typed matrix T_+, the reflection map R, and their extension shears on a substantial real-frequency band using the partial-jet system (7).

8 Exact threshold structure

The threshold $\omega = 0$ is excluded from the preceding holomorphic genericity argument and must be treated separately. Several exact facts can nevertheless be established before a uniform low-frequency scattering estimate.

For

$$D = \frac{r-2}{r}\partial_r, \qquad V_2 = \frac{6(r-2)(r-1)}{r^4}, \qquad V_1 = \frac{6(r-2)}{r^3},$$

consider $(D^2 - V_s)\phi = 0$.

Proposition 8.1 (Exact scalar threshold nonresonance)**.** *The horizon-normalized regular zero modes are*

$$\phi_{0,2}(r) = \frac{r^3}{8}, \qquad \phi_{0,1}(r) = \frac{r^2(2r-3)}{4}.$$

Independent solutions are, up to nonzero constant multiples,

$$\widetilde{\phi}_{0,2} = \frac{r^3}{32}\log\frac{r-2}{r} + \frac{r^2}{16} + \frac{r}{16} + \frac{1}{12} + \frac{1}{8r},$$

and

$$\widetilde{\phi}_{0,1} = \frac{3r^2(2r-3)\log\frac{r-2}{r} + 12r^2 - 6r - 2}{24}.$$

The regular modes grow as nonzero multiples of r^3. The decaying companions behave respectively as $-1/(5r^2) + O(r^{-3})$ and $-1/(10r^2) + O(r^{-3})$, but are logarithmically singular at the horizon. Consequently neither scalar factor has a bounded zero-energy resonance.

Proof. Direct substitution gives

$$(D^2 - V_s)\phi_{0,s} = 0, \qquad \phi_{0,s}(2) = 1,$$

and verifies the two reduction-of-order companions. Horizon regularity selects the polynomial solution in each scalar problem; neither selected solution is bounded at infinity. Conversely, the solutions that decay at infinity contain $\log(r-2)$. This exhausts each two-dimensional scalar solution space. □

The exact reduced projective cocycle also exposes the relation between the three factors:

$$\mathcal{I}_{\mathrm{red}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega} = \frac{2i}{5\omega}(V_1 - V_2) + \frac{i\omega(r-2)(2r+3)}{5r^4}. \tag{17}$$

Thus its singular threshold term is exactly the difference of the spin-one and spin-two potentials. On $\phi_{0,2}$ the leading source is

$$\frac{3i}{10\omega}\frac{r-2}{r}.$$

It has the elementary spin-two primitive

$$(D^2 - V_2)p = \frac{r-2}{r}, \qquad p = -\frac{r^2}{4} - \frac{r}{4} - \frac{1}{3} - \frac{1}{2r}. \tag{18}$$

The horizon-fixed choice

$$p_H = p + \frac{25}{12}\phi_{0,2}$$

satisfies $p_H(2) = 0$.

Remark 8.2 (Open matching estimate). Formal two-region matching to the far $\ell = 2$ Bessel equation predicts

$$|A_{\text{in},2}| \sim |A_{\text{out},2}| \sim \frac{15}{16\omega^3}, \qquad |A_{\text{in},1}| \sim |A_{\text{out},1}| \sim \frac{15}{4\omega^3},$$

and therefore

$$|\det T_+| \sim \frac{3375}{1024}\omega^{-9}.$$

The primitive (18) further suggests $b/a^2 = O(\omega^2)$ in the compatible intrinsic normalization. These asymptotics are not used as theorems here. A two-region Volterra remainder controlling the Schwarzschild tail and an endpoint-compatible extension normalization are still required before one may infer a punctured zero-frequency interval of outgoing invertibility or a weighted multiplier bound.

9 Separated-mode stability

In the Fourier convention used here, exponentially growing modes lie in the lower half ω-plane. For the scalar spin-one and spin-two Regge–Wheeler equations, an ingoing-horizon/outgoing-infinity solution in that half-plane decays at both ends. Their real, nonnegative potentials give the standard energy identity, forcing the scalar solution to vanish.

Theorem 9.1 (No growing axial separated mode)**.** *The complete axial $\ell = 2$ six-state system has no nonzero separated solution satisfying the future-horizon-regular and pure-outgoing-infinity conditions in the growth half-plane.*

Proof. The boundary maps preserve the declared complex-frequency classes, including regular replacements at the apparent frame events. Apply Lemma 6.1: the spin-one projection vanishes by the scalar energy identity, then the carrier spin-two quotient vanishes, and finally the metric spin-two factor vanishes. □

This is separated-mode stability. It does not supply limiting absorption, uniform resolvent bounds, threshold control, decay, or bounded time evolution. The extension ratios in (9) can still generate non-normal amplification.

10 A certified intrinsic connection exceptional point

At a simple spin-two quasinormal frequency ω_n, assume the spin-one factor is a local unit. Analytic elimination of that unit reduces the repeated block to

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \qquad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0. \tag{19}$$

Proposition 10.1 (Local Smith dichotomy)**.** *If $b(\omega_n) \neq 0$, the full 3×3 connection has Smith valuations*

$$(0, 0, 2),$$

rank two at the QNM, geometric multiplicity one, and one length-two root chain. If $b(\omega_n) = 0$, the valuations are

$$(0, 1, 1),$$

the rank is one, and there are two independent root vectors.

Proof. Work in the local discrete valuation ring with $z = \omega - \omega_n$. In the first case b is a unit, so the ideal of 2×2 minors is the unit ideal while $\det T$ has valuation two. The invariant factors are therefore $(0, 0, 2)$. In the second case $b = ag$, every 2×2 minor is divisible by a, while the minor af has valuation one; hence $(0, 1, 1)$. $\square$

The selector can be computed projectively. Transport the horizon-regular and infinity-outgoing scalar lines to a common matching section and use a Riccati chart $v = Y_2/Y_1$. With

$$\delta(\omega, \tau) = v_\infty(\omega, \tau) - v_H(\omega, \tau),$$

the scalar Evans coefficient has the form $a = g\delta$, where g is an analytic unit. At a simple QNM,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n)}{\delta_\omega(\omega_n)} = -\omega_n'(0). \tag{20}$$

Theorem 10.2 (Certified axial connection EP2)**.** *There is a radius-*10^{-7} *disk centered at*

$$\omega_c = -0.3736716844180418 + 0.0889623156889357\, i$$

that contains exactly one simple spin-two quasinormal zero in the declared $e^{+i\omega t}$ *convention. On that disk:*

1. *the validated projective Evans contour has winding* $+1$*;*
2. δ_ω *excludes zero;*
3. *the intrinsic tangent satisfies*

$$\delta_\tau \in [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

   and therefore excludes zero;

4. *the spin-one projective mismatch has modulus greater than* $1/2500$.

Consequently the complete axial connection has Smith valuations $(0, 0, 2)$*, algebraic multiplicity two, geometric multiplicity one, and one length-two connection root chain at the unique enclosed QNM.*

The exceptional point is intrinsic: the rational self-extension is not a frequency, mass, or angular-eigenvalue tangent modulo admissible rational gauge. What is certified is a pole of order two of the inverse *connection* block. To infer the rank-one principal part

$$-\frac{\beta_n}{\alpha_n^2} \frac{u_n \otimes \widetilde{u}_n}{(\omega - \omega_n)^2}$$

for the physical differential resolvent, one must still construct an analytic Fredholm realization with the QNM endpoint conditions and prove graph-norm boundedness of the extension.

11 Polar endpoint reach

The polar carrier includes trace terms:

$$\delta B_{ab} = \frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} - \frac{1}{6}\nabla_a\nabla_b\psi - \frac{1}{12}g_{ab}\Box\psi. \tag{21}$$

On the Bianchi-constrained polar $\ell = 2$ system the operator is traceless and has a one-function linearized conformal-gauge direction. On a traceless slice the horizon residue analysis yields, after quotienting the regular conformal direction, a two-parameter physical ingoing-analytic carrier family.

Corollary 11.1 (Parity-complete local endpoint nonselection)**.** *At the linear $\ell = 2$ mode level, both axial and polar Ricci carriers have nonzero future-horizon-regular data and propagate on the Einstein characteristics with admissible leading outer falloff. The tested local endpoint diagnostics therefore fail to select the Einstein image in either parity. The global populated Krein and connection theorems of this paper are axial only.*

12 Logical map and evidence states

The core structure is

$$\begin{tikzcd} 0 \arrow[r] & M_{\mathrm{RW}}^{(2)} \arrow[r] & M_{\mathrm{Bach}}^{\mathrm{axial}} \arrow[r, "\delta\,\mathrm{Ric}"] & M_{\mathrm{car}} \arrow[r] & 0 \\ & \cup & & \cap & \\ & F_1 \;\subset & F_2 \;\subset & F_3 = M_{\mathrm{axial}}, & \end{tikzcd}$$

with successive factors $(2, 2, 1)$. For real frequency, horizon data $h \in \mathbb{C}^3$ maps to

$$\begin{tikzcd} & h \arrow[dl, "T_-"'] \arrow[dr, "T_+"] & \\ \mathscr{I}^-_{\mathrm{in}} & & \mathscr{I}^+_{\mathrm{out}}, \end{tikzcd}$$

and the action current supplies

$$H_{\mathcal{H}^+} + T_+^\dagger G_+ T_+ - T_-^\dagger G_- T_- = 0.$$

Statement	Evidence state	Boundary
Ricci–Bach composition and exact sequence	Exact symbolic theorem	Schwarzschild, linearized, declared mode classes
RW/RW/spin-one filtration and partial jet	Exact symbolic theorem with independent certificate	Axial $\ell = 2$
Incoming factor Gram and $T_- \in GL(3, \mathbb{C})$	Exact plus computer-assisted theorem	Every real $\omega > 0$, $M = 1$ normalization
Outgoing population and pseudo-isometry	Validated interval theorem	I_*; genericity off a locally finite set
Exact scalar threshold nonresonance	Exact symbolic theorem	No punctured zero-free scattering interval claimed
No growing separated mode	Analytic filtered energy theorem	Axial $\ell = 2$, declared complex boundary classes
Connection Smith valuations $(0, 0, 2)$	Validated projective Evans and local-unit theorem	One enclosed simple damped QNM
Second-order Green-resolvent pole	Open conditional corollary	Requires analytic Fredholm realization
Time-domain boundedness and decay	Open	Requires threshold, limiting-absorption and resolvent bounds

13 Conclusions

The additional pure-Weyl carrier is not removed by the ordinary one-ended Lorentzian Schwarzschild endpoint conditions tested here. The result is stronger than a local degree-of-freedom count:

- the carrier belongs to an exact non-split filtered module;
- it enters the current derived from the Weyl action;
- the complete incoming flux space is nondegenerate and indefinite;
- every incoming direction is populated by a future-horizon-regular solution at every positive real frequency;
- outgoing population holds on a certified band and generically on the positive axis;
- the full filtered system has no growing separated mode;
- one physical spin-two QNM is a defective connection-level double root.

Thus negative classical flux directions coexist with separated-mode stability. The remaining threat to time-domain control is quantitative resolvent or pseudospectral growth, not an exponentially growing separated frequency.

The next two calculations are sharply defined. First, compute the explicit typed T_+ and reflection matrix over a substantial real-frequency band using the correlated partial-jet transport. Second, construct the analytic Fredholm QNM realization needed to promote Theorem 10.2 to a second-order Green-resolvent pole.

A Typed endpoint frames and explicit Grams

The triangular factor frame and the diagonal signature frame serve different purposes. After normalizing the repeated-spin-two hyperbolic plane, write

$$\mathcal{B}_0 = (E, R, S), \qquad J_0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The corresponding signature frame is

$$P = \frac{E+R}{\sqrt{2}}, \qquad N = \frac{E-R}{\sqrt{2}}, \qquad S = S,$$

with

$$J_\sigma = \mathrm{diag}(1, -1, -1), \qquad C = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad C^\dagger J_0 C = J_\sigma.$$

If M_0 is written in the factor frame, then

$$M_\sigma = C^{-1} M_0 C.$$

For

$$M_0 = \begin{pmatrix} m & x & y \\ 0 & m & z \\ 0 & 0 & n \end{pmatrix},$$

one finds

$$M_\sigma = \begin{pmatrix} m + x/2 & -x/2 & (y+z)/\sqrt{2} \\ x/2 & m - x/2 & (y-z)/\sqrt{2} \\ 0 & 0 & n \end{pmatrix}.$$

Thus triangularity is a statement about the filtration frame, not the signature frame. In particular a nonzero extension entry x produces both upper and lower off-diagonal entries after signature normalization.

In the original incoming order $(XI0, XI1, EI0)$, the normalized action Gram is

$$G_-^{\mathrm{raw}} = \begin{pmatrix} \frac{576}{5\omega} & \frac{96i}{5} & \frac{384\omega}{5} \\ -\frac{96i}{5} & -\frac{144\omega}{5} & -\frac{192i\omega^2}{5} \\ \frac{384\omega}{5} & \frac{192i\omega^2}{5} & 0 \end{pmatrix}. \tag{22}$$

The factor shear and orthogonal quotient lift reduce (22) to (11).

For reference, in the original outgoing order $(XI2, XI3, EI2)$, the normalized future Gram is

$$G_+^{\mathrm{raw}} = \begin{pmatrix} \frac{384(-512\omega^6+640\omega^4-278\omega^2+39)}{5\omega} & \frac{12288i\omega^4-3072\omega^3-9984i\omega^2}{5}+192\omega+384i & \frac{-1536i\omega^2+384\omega+768i}{5} \\ \frac{-12288i\omega^4-3072\omega^3+9984i\omega^2}{5}+192\omega-384i & -\frac{768\omega^3}{5}+48\omega & \frac{96\omega}{5} \\ \frac{1536i\omega^2+384\omega-768i}{5} & \frac{96\omega}{5} & 0 \end{pmatrix}. \tag{23}$$

The full physical Grams are $\pi\alpha_{\mathrm{W}} G_\pm^{\mathrm{raw}}$. On $[1/2, 3/4]$, exact rational/Sturm bounds give

	$\max \lVert G_\pm \rVert_F^2$	$\max \lVert G_\pm^{-1} \rVert_F^2$
G_-	1429056/25	7025/65536
G_+	10389888/25	19825/65536.

These estimates give uniform equivalence of the finite-band coefficient norms without inferring inverse bounds from determinants alone.

A.1 The quotient metric is intrinsic

Let K be the repeated-spin-two plane and M_1 the spin-one quotient. Because $G_-|_K$ is nondegenerate,

$$\pi_1 : K^{\perp_{G_-}} \longrightarrow M_1$$

is an isomorphism. If $s_\perp$ denotes its inverse, then

$$g_1(q_1, q_2) = G_-(s_\perp q_1, s_\perp q_2)$$

is independent of every arbitrary lift. In a nonorthogonal adapted frame it is the Schur complement

$$g_1 = G_{11} - G_{1K} G_{KK}^{-1} G_{K1}.$$

The value $g_1 = -32\pi\alpha_{\mathrm{W}}/(15\omega)$ is therefore a quotient-theoretic statement, not a sign attached to one chosen Witt vector.

B Krein scattering identities

In either typed frame define the Krein adjoint

$$M^\sharp = J^{-1} M^\dagger J.$$

After separate endpoint normalizations, the one-sided conservation law is

$$R^\sharp R + A^\sharp A = I.$$

The horizon defect operator and its Hermitian form are

$$\mathfrak{D}_H = I - R^\sharp R = A^\sharp A, \qquad D_H = J - R^\dagger J R = A^\dagger J A.$$

On the certified cell $A = T_-^{-1}$ is invertible and the horizon Gram has inertia $(1, 2, 0)$. Hence

$$\mathfrak{D}_H \in GL(3, \mathbb{C}), \qquad 1 \notin \sigma(R^\sharp R), \qquad \operatorname{inertia} D_H = (1, 2, 0).$$

The defect is nondegenerate but indefinite. It is not a positive absorption operator, and the conservation theorem supplies no inequality of the form "outgoing flux is at most incoming flux" on the complete Weyl trace space.

In canonical normalized frames $\det J = 1$, so

$$\det D_H = |\det A|^2 = |\det T_-|^{-2}.$$

Using

$$\det T_- = A_{\text{in},2}^2 A_{\text{in},1}$$

and the scalar Wronskians,

$$\det D_H = |A_{\text{in},2}|^{-4} |A_{\text{in},1}|^{-2}, \qquad 0 < \det D_H \leq 1.$$

This positive signed three-volume is the surviving determinant-level remnant of ordinary scalar absorption; the individual defect eigenvalues need not lie in $[0, 1]$.

An algebraic two-sided completion also follows. The range of $\mathscr{S} = (R, A)^T$ is nondegenerate with inertia $(1, 2)$ in a target of inertia $(2, 4)$. Its orthogonal complement has inertia $(1, 2)$. Appending a $J \oplus J$-orthonormal basis of that complement gives an element of $U(2, 4)$. This is only an algebraic completion; it does not construct the physical past-horizon channels.

C Projective Evans reduction and the certified disk

For a scalar companion system

$$Y_x = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} Y$$

use the projective coordinate $v = Y_2/Y_1$. It satisfies

$$v_x = A_{21} + (A_{22} - A_{11})v - A_{12}v^2.$$

Let v_H and v_∞ be the horizon-regular and infinity-outgoing lines transported to a common section. Their mismatch

$$\delta = v_\infty - v_H$$

obeys the exact scalar equation

$$\delta_x = \big[A_{22} - A_{11} - A_{12}(v_\infty + v_H)\big]\delta.$$

It follows that projective Evans functions at two matching sections differ by an analytic nonzero factor. Their zero divisors, logarithmic contour moments, and local velocity ratios are match-section independent.

If τ denotes the intrinsic partial-jet deformation, the tangent variables satisfy

$$(v_p)_x = F_v v_p + F_p, \qquad p \in \{\omega, \tau\},$$

where F is the Riccati right-hand side. At a simple zero,

$$\kappa_n = \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n)}{\delta_\omega(\omega_n)}.$$

This removes every homogeneous amplitude from the Smith selector.

On a boundary contour avoiding zeros,

$$\frac{1}{2\pi i} \oint \frac{\delta_\omega}{\delta}\, \mathrm{d}\omega$$

counts scalar QNMs, while

$$\frac{1}{2\pi i} \oint \frac{\delta_\tau}{\delta}\, \mathrm{d}\omega$$

is the sum of their intrinsic velocity classes. A projective chart change is a Möbius transformation. It multiplies δ by an analytic nonzero unit whenever both endpoint lines remain in the chart, so the closed-contour invariants are unchanged.

The certified disk in Theorem 10.2 was closed in two stages. First, validated projective panels excluded zero on the boundary and supplied a lifted argument change of 2π. Second, a local interval-Newton rail excluded zero from δ_ω and enclosed the unique simple root. The intrinsic tangent and the independent spin-one mismatch were then evaluated on the same disk. This separation prevents a numerical root approximation from being reused as its own simplicity or local-unit proof.

D Certificate authorities

The principal machine authorities are:

- `black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json`;
- `black_hole_programme/phase3/axial_rw_lx_triangular_preflight/certificate.json`;
- `black_hole_programme/phase3/axial_null_flux_gram/certificate.json`;
- `black_hole_programme/phase3/axial_incoming_extended_domain_audit/certificate.json`;
- `black_hole_programme/phase3/axial_boundary_devissage_no_growth/certificate.json`;
- `black_hole_programme/phase3/axial_outgoing_population_cell_half_v1/certificate.json`;
- `black_hole_programme/phase3/axial_global_finite_flux_channel_classification_v3/certificate.json`;
- `black_hole_programme/phase4/axial_threshold_exact_structure_v1/certificate.json`;

- `black_hole_programme/phase3/axial_qnm_projective_evans_contour_completion/full_contour_winding_v1/certificate.json`;
- `black_hole_programme/phase3/axial_qnm_projective_evans_contour_completion/local_selector_v1/certificate.json`;
- `black_hole_programme/phase3/axial_qnm_spin_one_local_unit_v1/certificate.json`;
- `black_hole_programme/certificates/BH2B_POLAR_REACH.json`.

Each numerical theorem has a separately implemented verifier and mutation tests. Reproduction commands and content hashes are recorded in the generated claim map and receipt accompanying this paper.

E What this paper does not establish

It does not establish:

- explicit T_+ entries or reflection amplitudes over a substantial frequency band;
- absence of isolated positive-real reflection zeros;
- a punctured threshold interval of outgoing invertibility or the formal low-frequency Jost and extension-ratio asymptotics in Section 8;
- a complete asymptotically flat metric/tetrad phase space and charge algebra;
- a polar global connection theorem;
- an analytic Fredholm realization of the damped QNM boundary problem;
- a physical second-order Green-resolvent pole or generalized ringdown term;
- limiting absorption, uniform resolvent bounds, threshold control, bounded time evolution, decay, or completeness;
- a positive quantum metric, particles, a BRST physical Hilbert space, CPT, or unitarity.

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